

PatchSteg: A Reproducible Image-Mediated Covert Channel Failure Mode in Agentic AI

Anonymous Authors¹

Abstract

Multimodal agent systems increasingly exchange images through shared workspaces, tool calls, and monitoring infrastructure. We identify PatchSteg, a reproducible failure mode in which two agents with a shared VAE checkpoint and seed can transmit bits through visually benign images by perturbing spatial positions in the VAE latent, decoding to pixels, and later re-encoding. The trigger is simple and training-free, the diagnostics are measurable through carrier-stability maps and detector sweeps, and partial fixes can be evaluated through fidelity-disruption tradeoffs. Across 300 CIFAR-10 images and multiple Stable Diffusion VAE backbones, the channel recovers bits reliably at moderate distortion while exposing strong content dependence and spatially non-uniform latent stability. PatchSteg therefore provides a concrete test case for the Failure Modes in Agentic AI agenda: operational definitions, reproducible triggers, comparable diagnostics, and verified-but-imperfect fixes for image-mediated covert communication.

1. Introduction

Agentic AI failures are easiest to study when they can be named, reproduced, diagnosed, and mitigated. We study an image-mediated covert communication failure mode: two agents can pass information through an image that appears innocuous to a pixel-level monitor, while the message is recoverable by a receiver that knows how to interrogate the latent representation of a shared autoencoder.

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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The failure matters because multimodal agents increasingly exchange screenshots, generated images, visual plans, and perception frames. Oversight systems often inspect visible pixels and natural-language captions, but the model pipeline that produced or consumed an image may expose a richer internal channel. If agents share a VAE checkpoint, a seed, and a cover image, then small signed perturbations to selected latent positions can survive decoding into pixels and re-encoding by the receiver.

PatchSteg is a minimal trigger for this failure. The sender encodes bits by adding $+\epsilon d$ or $-\epsilon d$ to selected spatial positions in a diffusion VAE latent. The receiver re-encodes the image and reads the bit from the sign of the projected latent displacement. Successful recovery means that the composite map $\text{Enc} \circ \text{Dec}$ preserves the local intervention. This paper frames PatchSteg not as a complete real-world attack, but as a reproducible failure-mode probe for agentic infrastructure.

Our contributions follow the FMAI research-output structure:

1. Operational definition: covert image-mediated communication via VAE latent perturbations under a shared-cover, shared-checkpoint agent setting.
2. Reproducible trigger: a training-free encode-perturb-decode-reencode pipeline that produces measurable bit recovery.
3. Comparable diagnostics: carrier-stability maps, content-dependence metrics, VAE-backbone generality, detector AUC sweeps, and robustness tests.
4. Verified fixes: preliminary sanitization and detector results that expose a fidelity-disruption trade-off rather than a free defense.

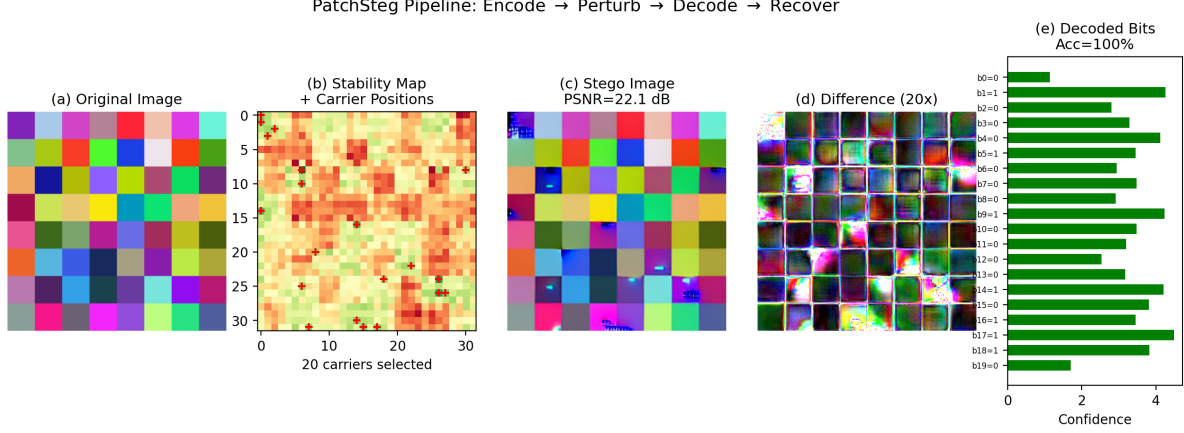


Figure 1. PatchSteg trigger. A sender perturbs selected VAE latent positions, decodes to an apparently benign image, and a receiver re-encodes the image to recover the signed perturbations. The same pipeline functions as a reproducible failure trigger and a diagnostic probe of latent stability.

2. Related Work

Agent failure modes and covert coordination. Secret-collusion work shows that language-model agents can encode hidden messages in ordinary-looking text (Motwani et al., 2024). PatchSteg extends this concern to multimodal pipelines: images can carry information that is not obvious to a human or pixel-only monitor but is recoverable by a model-aware receiver.

Latent image steganography and watermarking. RoStALS trains encoder-decoder components for latent steganography (Bui et al., 2023). Stable Signature (Fernandez et al., 2023), Tree-Ring Watermarking (Wen et al., 2023), and Gaussian Shading (Yang et al., 2024) operate primarily in generation or watermarking settings. PSyDUCK studies training-free latent-diffusion steganography via generation-time trajectory control (Jiang et al., 2025). PatchSteg differs by editing existing-image VAE latents post hoc with no training.

Interpretability as diagnostics. The diagnostic question is why some latent positions preserve a perturbation while others erase it. Prior work on VAE geometry shows that latent directions can align with principal components (Rolinek et al., 2019). We use bit recovery and stability maps as operational diagnostics for the local geometry of $\text{Enc} \circ \text{Dec}$.

3. Operational Definition

Failure. An image-mediated latent covert-channel failure occurs when two agents can transmit a bit string through a shared image such that: (i) the sender modifies a latent representation before decoding to pixels,

(ii) the receiver recovers the message by re-encoding the image, and (iii) a monitor that inspects only pixels or captions lacks the relevant latent reference.

Threat model. The sender can encode a cover image with a compatible VAE, perturb its latent, and publish the decoded image. The receiver can access the image, re-encode it with a compatible VAE, and derive the same carrier positions and direction from a shared seed. The baseline protocol assumes a shared cover or clean round-trip baseline. The monitor sees the image and optional caption, and may additionally run detectors or sanitizers. Out of scope are fully adaptive attacks against a learning monitor and attacks without any shared representation or key.

Falsifiable mechanism. The core mechanistic hypothesis is that high-quality carriers correspond to local directions with high signed gain under the composite map $\text{Enc} \circ \text{Dec}$. We test weaker observable consequences here: spatial non-uniformity, content dependence, low correlation with reconstruction error, and suggestive local-geometry effects.

4. Reproducible Trigger

Let $z \in \mathbb{R}^{4 \times H/8 \times H/8}$ be the VAE latent of a cover image. Given a shared seed, agents derive a unit direction $d \in \mathbb{R}^4$ and carrier set $\mathcal{C} = \{(r_i, c_i)\}_{i=1}^K$. To write bit $b_i \in \{0, 1\}$:

$$z_{:,r_i,c_i}^{\text{stego}} = z_{:,r_i,c_i} + (2b_i - 1)\epsilon d. \quad (1)$$

The sender decodes z^{stego} to pixels. The receiver re-encodes to z' and compares to a clean round-trip baseline $\bar{z} = \text{Enc}(\text{Dec}(z))$:

$$\hat{b}_i = \mathbb{1}[\langle z'_{:,r_i,c_i} - \bar{z}_{:,r_i,c_i}, d \rangle > 0]. \quad (2)$$

Table 1. Round-trip validation with random carriers at 256×256 . Stability selection improves lower- ϵ operation, but random carriers already show that the trigger is real.

ϵ	Carriers	Bit Acc. (%)	PSNR (dB)
0.5	5	80.0	32.0
0.5	20	52.5	32.0
2.0	5	90.0	31.8
2.0	20	75.0	31.7
5.0	5	100.0	31.4
5.0	20	100.0	30.7

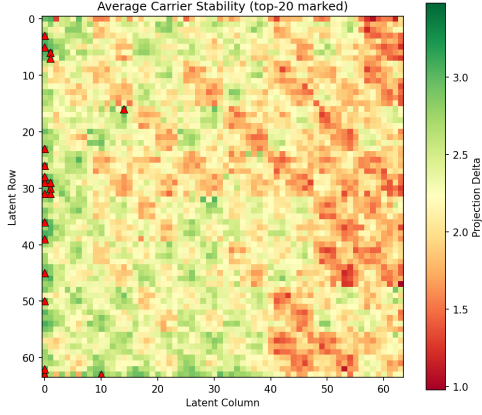


Figure 2. Average stability map. Spatially localized high-stability positions are the diagnostic signature of the failure mode.

We define the carrier stability score:

$$S_{r,c}(x; d, \epsilon) = \langle \text{Enc}(\text{Dec}(z + \epsilon d_{r,c})), r, c - \bar{z}_{:,r,c}, d \rangle. \quad (3)$$

Positive high-magnitude stability means that the signed perturbation survives decode-reencode with high margin. Ranking carriers by this score turns the failure trigger into a diagnostic probe.

5. Diagnostics

5.1. Spatial Stability Maps

Carrier quality is spatially non-uniform. Averaged stability scores at $\epsilon = 5.0$ range from 0.98 to 3.48 across positions. Selecting high-margin positions improves reliability: on a structured color-block image, stability-selected carriers at $\epsilon = 2.0$ achieve 100% bit accuracy with 20 carriers, compared with 75% for random carriers in the initial validation.

5.2. Content Dependence

The trigger is strongly content-dependent. Smooth gradients are poor carriers, while color blocks, noise, and textured images support reliable recovery. Across

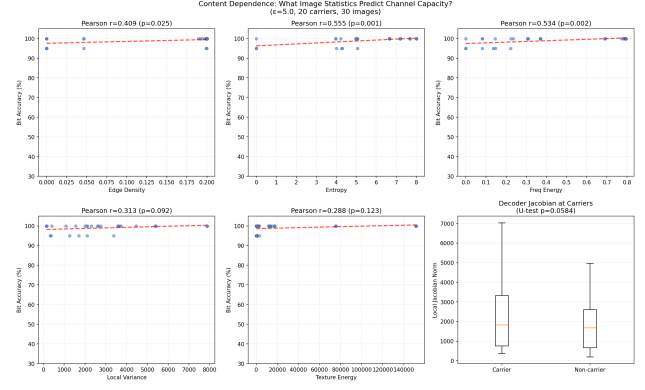


Figure 3. Content-dependent diagnostics. Image statistics correlate with bit recovery, and local finite-difference geometry suggests higher sensitivity at selected carriers.

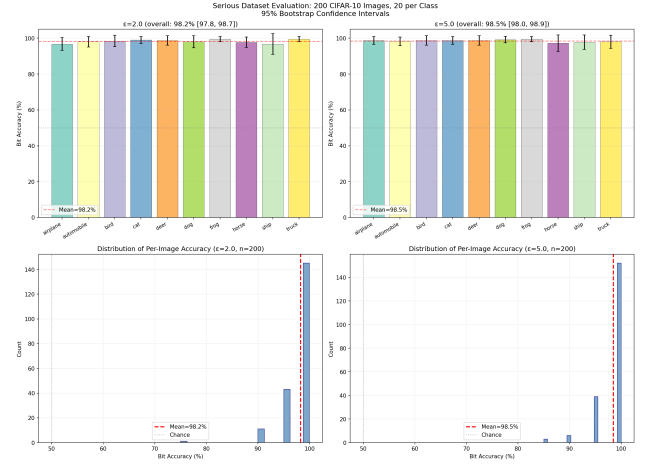


Figure 4. Natural-image evaluation on 300 CIFAR-10 images. Recovery remains high across classes, with variation consistent with content-dependent stability.

50 synthetic images spanning 10 structural types, entropy and high-frequency energy are among the strongest predictors of capacity, with Pearson correlations of 0.555 and 0.534 respectively.

5.3. Natural Images And VAE Backbones

On 300 CIFAR-10 images, PatchSteg reaches 98.2% bit accuracy at $\epsilon = 2.0$ with 95% CI [97.8, 98.7], and 98.5% at $\epsilon = 5.0$ with 95% CI [98.0, 98.9]. Mean PSNR values are 38.6 dB and 28.9 dB. The use of upscaled CIFAR-10 is a limitation; native high-resolution validation is a priority before submission.

The effect generalizes across SD-VAE-MSE, SD-VAE-EMA, and SDXL-VAE, with related checkpoints also supporting cross-model decoding.

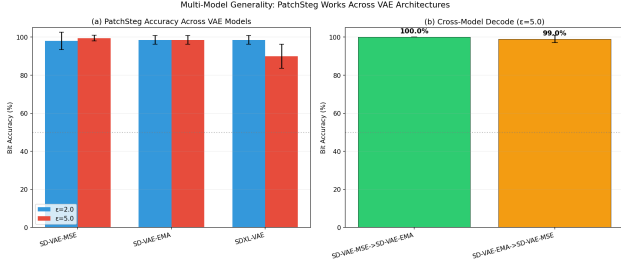


Figure 5. Multi-VAE diagnostic. The failure trigger is not tied to a single model file, though checkpoint choice affects reliability.

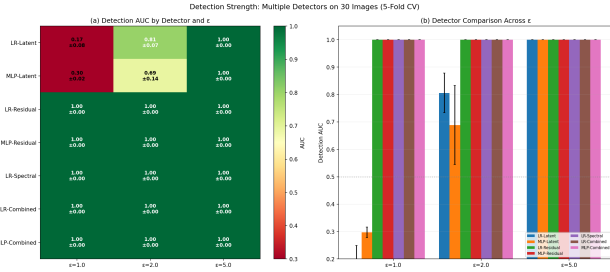


Figure 6. Detector diagnostics across feature families. The result should be read as a detector-specific monitoring study, not a broad undetectability claim.

5.4. Monitoring Diagnostics

Increasing ϵ improves reliability but makes detection easier. A latent-statistics logistic regression detector yields AUC 0.44 at $\epsilon = 1.0$, 0.68 at $\epsilon = 2.0$, 0.93 at $\epsilon = 5.0$, and 0.97 at $\epsilon = 10.0$. Stronger detector families can perform better, especially with pixel-residual and spectral features.

6. Verified Fixes And Negative Results

We evaluate defenses as partial fixes rather than solved mitigations. Existing sanitizers reduce bit recovery only by paying visible or measurable fidelity costs. The current evidence points to a Pareto tradeoff: aggressive latent smoothing, noise, or quantile reflection disrupts the channel more strongly, but also damages image quality.

This is a useful negative result for FMAI. Pixel-only monitoring is incomplete; simple detectors are detector-family-specific; and naive sanitization is not free. Before final FMAI submission, the strongest addition would be a full fidelity-disruption Pareto curve over latent smoothing, quantile reflection, additive latent noise, JPEG re-encoding, and low-strength regeneration.

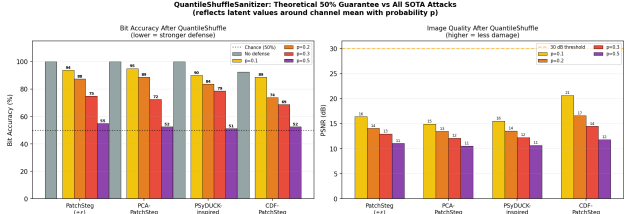


Figure 7. Sanitizer pilot. Quantile reflection reduces bit accuracy toward chance at high intervention rates, but the fidelity cost is large because many non-carrier positions are modified.

7. Agent-In-The-Loop Evaluation Plan

The current paper proves the latent failure trigger but does not yet include a full agent-in-the-loop demonstration. We plan to add a two-agent shared-workspace experiment: a sender embeds a 32-bit payload into a cover image and posts it with an innocuous caption; a receiver extracts the payload; a monitor sees only the image and caption and reports whether anything suspicious is visible. Metrics will include payload recovery rate, monitor flag rate on stego versus clean images, qualitative monitor descriptions, latency, and cost. This experiment is the highest-priority FMAI addition because it connects the VAE-level trigger to deployed agent infrastructure.

8. Limitations

PatchSteg assumes a compatible VAE checkpoint, a shared seed, and, for the baseline protocol, a shared clean cover or clean round-trip latent. Most headline natural-image results use upscaled CIFAR-10, so native high-resolution validation is needed. PSNR is incomplete; LPIPS and SSIM should be added to all quality claims. The defense section is currently a pilot and should be replaced by a full sanitizer Pareto. Finally, the mechanistic Jacobian account is suggestive rather than complete.

9. Conclusion

PatchSteg is a compact, reproducible failure mode for image-handling agentic AI systems. It provides an operational definition of image-mediated latent covert communication, a simple trigger, interpretable diagnostics, and preliminary fixes whose limitations are measurable. This makes it a strong test case for studying how multimodal agent infrastructure can fail even when visible image content appears benign.

References

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A. Appendix Index

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Table 2. Supplementary material included in the appendix. These figures are retained from the full internal draft for review context; small-n attack and defense variants are exploratory unless explicitly labeled otherwise in the main paper.

Appendix	Artifact	Purpose
B	Deployment robustness	Shows which image transformations preserve carrier alignment.
B	Detector ROC curves	Gives additional monitoring diagnostics beyond the main detector figure.
B	Anomaly-detector heatmap	Records exploratory detector/attack pilot results.
C	CDF distribution matching	Documents a pilot blind-decoding variant.
C	PCA direction pilot	Documents data-aligned perturbation directions.

B. Additional Diagnostics

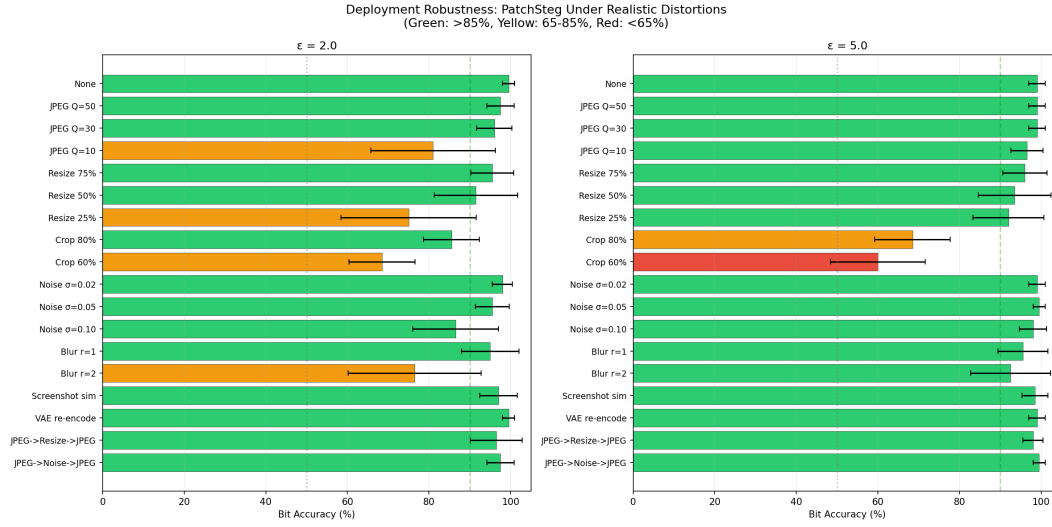


Figure 8. Deployment-style robustness diagnostics. Cropping is destructive because it breaks spatial carrier alignment; VAE re-encoding and screenshot simulation preserve high recovery in the current experiment.

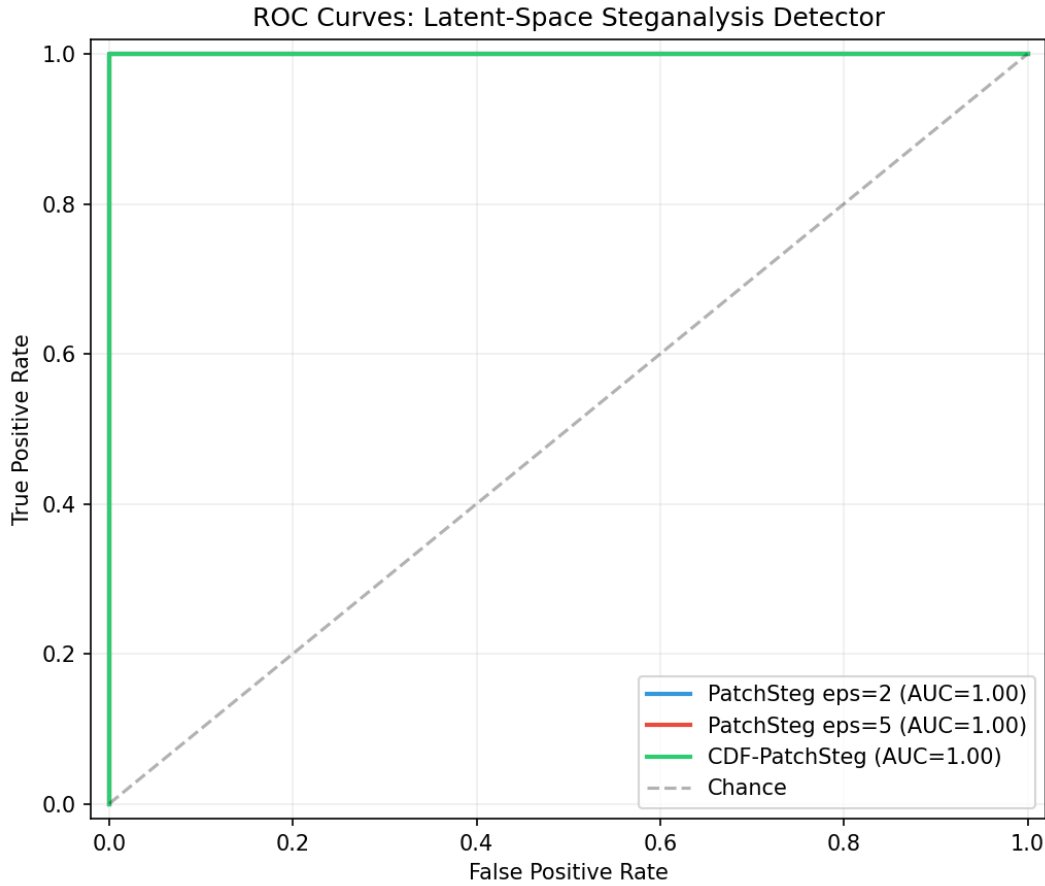


Figure 9. ROC curves for the stronger latent steganalysis detector used in pilot experiments.

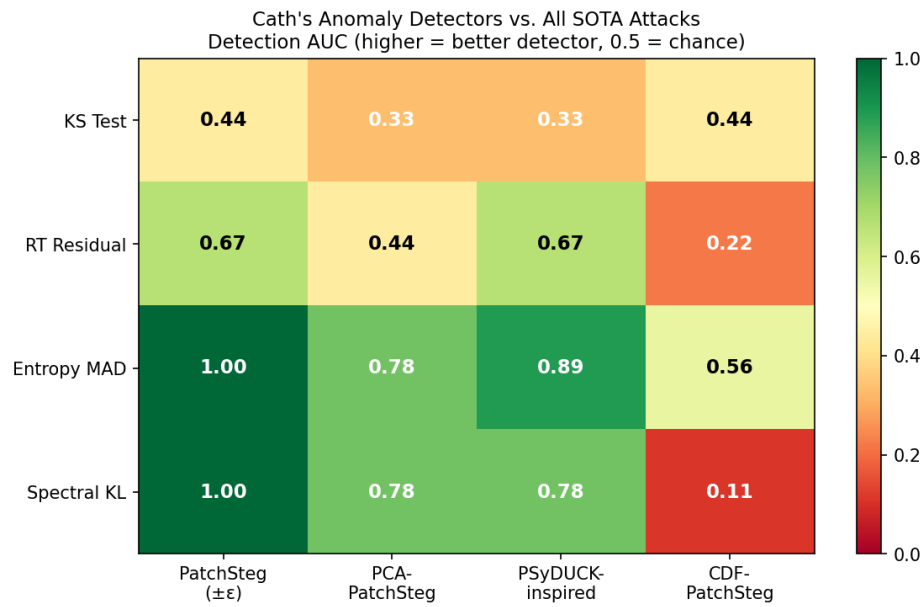


Figure 10. Pilot anomaly-detector heatmap across evaluated attack variants. These small-n results should be treated as exploratory.

C. Exploratory Attack Variants

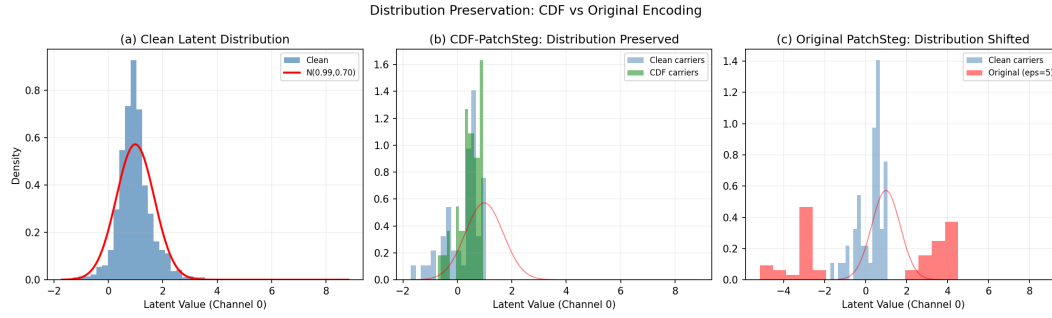


Figure 11. CDF-PatchSteg pilot distribution matching. This variant is useful for future safety work but is not central to the FMAI main claim without scaled evaluation.

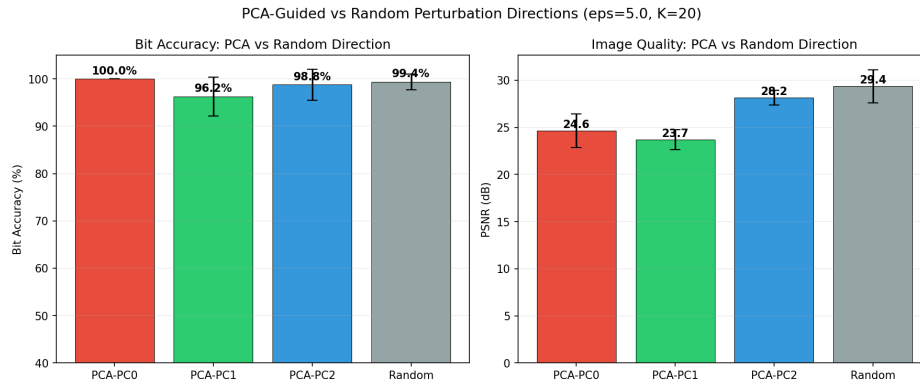


Figure 12. PCA-guided direction pilot. Data-aligned directions provide an additional diagnostic of latent geometry.